

Python Data Analysis Assignment – NumPy and Pandas

Project Overview

This project focuses on performing basic data analysis using NumPy and Pandas in Python.

The project covers:

- ☐ NumPy Array Operations
- ☐ Pandas Series Operations
- ☐ Pandas DataFrame Operations

The main purpose of this project is to understand how to create, inspect, slice, manipulate, filter, and analyze data using Python.

Objectives

- ☐ To create and work with NumPy arrays.
- ☐ To understand array shape, data type, and number of elements.
- ☐ To perform mathematical operations using NumPy.
- ☐ To perform indexing and slicing.
- ☐ To create and manipulate Pandas Series.
- ☐ To understand loc and iloc.
- ☐ To apply Boolean filtering.
- ☐ To create and analyze a Pandas DataFrame.
- ☐ To perform filtering, grouping, and aggregation.
- ☐ To modify and delete rows and columns.

Technologies and Libraries Used

- ☐ Python
- ☐ NumPy
- ☐ Pandas
- ☐ Jupyter Notebook

1. NumPy Array Operations

1.1 Creating a 1D NumPy Array

A 1D NumPy array named `temperature_w1` was created using the daily average temperature values for Week 1.

Temperature values:

22.5, 25.3, 20.8, 23.4, 26.1, 24.8, 21.9

1.2 Inspecting Array Properties

The shape, data type, and number of elements were inspected.

Shape: (7,)

Data Type: float64

Number of Elements: 7

1.3 Celsius to Fahrenheit Conversion

The temperature values were converted from Celsius to Fahrenheit using the formula:

$$\text{Fahrenheit} = (\text{Celsius} \times 9/5) + 32$$

The Fahrenheit temperatures obtained were:

72.5, 77.54, 69.44, 74.12, 78.98, 76.64, 71.42

1.4 Maximum, Minimum and Mean Temperature

Maximum Temperature: 26.1°C

Minimum Temperature: 20.8°C

Mean Temperature: 23.542857°C

2. NumPy Array Slicing and Indexing

2.1 First Three Days

The first three temperature values were extracted.

Result:

22.5, 25.3, 20.8

2.2 Weekend Temperatures

The last two temperature values were extracted as weekend temperatures.

Result:

24.8, 21.9

2.3 Middle Three Days

The middle three temperature values were extracted.

Result:

20.8, 23.4, 26.1

3. Creating a 2D NumPy Array

A 2D NumPy array named temperatures was created.

Week 1:

22.5, 25.3, 20.8, 23.4, 26.1, 24.8, 21.9

Week 2:

19.2, 22.5, 21.3, 24.0, 23.5, 22.8, 20.1

4. Inspecting the 2D Array

Shape: (2, 7)

Data Type: float64

Total Number of Elements: 14

The array contains two rows and seven columns.

5. Slicing the 2D Array

5.1 Temperatures for Each Week

Week 1 temperatures were extracted from the first row.

Week 2 temperatures were extracted from the second row.

5.2 Weekend Temperatures for Both Weeks

Week 1 Weekend:

24.8, 21.9

Week 2 Weekend:

22.8, 20.1

6. Pandas Series

6.1 Creating a Pandas Series

A Pandas Series named marks was created using student marks and custom rank labels.

Rank1: 95

Rank2: 92

Rank3: 89

Rank4: 85

Rank5: 80

7. Pandas Series Indexing and Slicing

7.1 Accessing the First Rank

The first-ranked student's mark was accessed using integer position indexing.

Result:

95

7.2 Retrieving the Top Three Ranks

The top three ranks were retrieved using the loc accessor.

Rank1: 95

Rank2: 92

Rank3: 89

7.3 Retrieving the Third Rank

The third-ranked student's mark was retrieved using iloc.

Result:

89

7.4 Filtering Marks Greater Than 90

A Boolean mask was applied to filter marks greater than 90.

Result:

Rank1: 95

Rank2: 92

8. Manipulating the Pandas Series

8.1 Modifying Rank1 Mark

The mark of Rank1 was changed from 95 to 100.

8.2 Removing Rank5

The entry corresponding to Rank5 was removed from the Series.

8.3 Calculating CGPA

CGPA was calculated by dividing each mark by 10.

Updated CGPA:

Rank1: 10.0

Rank2: 9.2

Rank3: 8.9

Rank4: 8.5

9. Pandas DataFrame

9.1 Creating the DataFrame

A Pandas DataFrame named transactions was created.

The DataFrame contains the following columns:

- ☐ TransactionID
- ☐ ProductCategory
- ☐ Region
- ☐ Amount

The DataFrame contains 10 transactions.

10. Data Exploration

10.1 Displaying Basic DataFrame Information

The DataFrame, shape, column names, data types, head, and tail were displayed.

Shape:

(10, 4)

Columns:

TransactionID, ProductCategory, Region, Amount

10.2 Displaying Selected Columns

Only the ProductCategory and Amount columns were displayed.

10.3 Retrieving the Last Three Columns

The last three columns were retrieved using the iloc accessor.

The columns were:

ProductCategory

Region

Amount

11. Filtering DataFrame Rows

Rows where Region was North and Amount was greater than 200 were filtered.

The matching transactions were:

TransactionID 103 – Amount 300

TransactionID 106 – Amount 250

TransactionID 110 – Amount 400

12. Value Counts

The number of transactions in each ProductCategory was calculated.

Electronics: 4

Clothing: 3

Furniture: 3

13. Unique Regions

The unique values in the Region column were identified.

North

South

East

West

14. Grouping and Mean Calculation

The DataFrame was grouped by Region and the mean Amount was calculated.

East: 375.0

North: 287.5

South: 250.0

West: 190.0

15. Manipulating the DataFrame

15.1 Modifying Transaction Amount

The Amount for TransactionID 102 was changed from 150 to 165.

15.2 Adding a Discount Column

A new column named Discount was created.

The Discount was calculated as 10% of Amount.

$\text{Discount} = \text{Amount} \times 0.10$

15.3 Removing TransactionID 109

The row corresponding to TransactionID 109 was removed.

15.4 Deleting the Discount Column

The Discount column was deleted from the DataFrame.

The final DataFrame contains:

TransactionID

ProductCategory

Region

Amount

16. Key Concepts Learned

Through this assignment, the following concepts were practiced:

- ☐ NumPy array creation
- ☐ 1D arrays
- ☐ 2D arrays
- ☐ Array shape
- ☐ Array data type
- ☐ Array size
- ☐ Mathematical operations
- ☐ Celsius to Fahrenheit conversion
- ☐ Maximum, minimum and mean
- ☐ Array indexing
- ☐ Array slicing
- ☐ Pandas Series creation
- ☐ Custom indexing
- ☐ loc and iloc
- ☐ Boolean masking
- ☐ Series modification
- ☐ Series deletion
- ☐ CGPA calculation
- ☐ DataFrame creation
- ☐ DataFrame exploration
- ☐ Column selection
- ☐ Row filtering
- ☐ Value counts
- ☐ Unique values
- ☐ GroupBy operation
- ☐ Mean aggregation
- ☐ DataFrame modification
- ☐ Adding columns
- ☐ Removing rows
- ☐ Deleting columns

17. Conclusion

This assignment provided practical experience in using NumPy and Pandas for data analysis.

NumPy was used to analyze weekly temperature data through array creation, mathematical operations, temperature conversion, indexing, slicing, and statistical calculations.

Pandas Series was used to analyze student marks using custom indexing, loc, iloc, Boolean filtering, modification, deletion, and CGPA calculation.

Pandas DataFrame was used to analyze transaction data through data exploration, column selection, filtering, value counts, unique values, grouping, mean calculation, row modification, column creation, row deletion, and column deletion.

Overall, this project helped to build a strong foundation in Python Data Analytics using NumPy and Pandas.

End of Project Documentation